

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	[Enter Team ID]
Project Title	HDI Predictor - Country Human Development Index Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This project automates the country human development index calculation and prediction process using machine learning. It analyses country socio-economic and demographic indicators to predict the HDI score and development tier. The solution reduces manual data wrangling effort, improves consistency, speeds up policy scenario analysis, and minimizes the risk of mathematical errors.

Project Overview	
Objective	Develop an intelligent machine learning system that predicts country Human Development Index (HDI) scores and development tiers based on socio-economic indicators with high accuracy.
Scope	The project includes data collection, preprocessing, feature engineering, model training, evaluation, and deployment using Flask. Multiple machine learning algorithms are compared to identify the best-performing model for real-time prediction.
Problem Statement	
Description	Policy makers and researchers analyze country development trends regularly. Manual calculation is time-consuming, inconsistent, and prone to mathematical errors, resulting in delayed policy evaluations and lagging static reports.
Impact	Automating the index prediction process improves scenario analysis speed, increases forecasting accuracy, reduces research timeline delays, minimizes manual calculation effort, and enhances user satisfaction.
Proposed Solution	
Approach	Historical country-year data is pre-processed and used to train Machine Learning models including Linear Regression, Random Forest, and Gradient Boosting. The best-performing model is integrated into a Flask web application for real-time prediction.
Key Features	• Automated HDI Score & Tier Prediction • High Accuracy using Machine Learning • Real-Time Prediction through Web Application • User-Friendly Interface • Fast Scenario-Simulation Process • Dockerized Container Deployment Support • Secure and Scalable Architecture

Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor (4+ Cores)
Memory	RAM specifications	8 GB RAM (Minimum), 16 GB Recommended
Storage	Disk space for datasets, trained models, and application files	512 GB SSD / 1 TB HDD
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data		
Data	Source, size, format	UNDP Human Development Index Dataset, CSV Format, ~4,500+ country-year records